LUNA ZEA-REDONDO, Ph.D.

Data Scientist | Computational Biology

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SUMMARY

Computational Biologist and Data Scientist with **8+ years** translating complex biological questions into data-driven insights. **PhD in Systems Biology** with expertise in **time-series analysis**, statistical modeling, and multi-omics data integration. With a strong experimental background, I excel at **designing end-to-end projects** and building **reproducible analytical workflows** in cross-functional environments. Strong **problem-solver**, eager to learn, and passionate about delivering insights that support **data-informed decision-making**.

PROFESSIONAL EXPERIENCE

Data Scientist | Computational Biology **Oct 2025 – present**
Berlin Institute for Medical Systems Biology (MDC-BIMSB) *Berlin, Germany*

- **Delivered** a **first-author multi-omics study** to final submission, owning project design, experimental execution, computational analysis and narrative development in alignment with senior stakeholders.
- **Led** the design and execution of **2 collaborative research initiatives** derived from my PhD research, coordinating delivery across **4 international cross-functional teams**.

Ph.D. Researcher | Computational Biology **Jan 2019 – Oct 2025**
Berlin Institute for Medical Systems Biology (MDC-BIMSB) *Berlin, Germany*

- **Designed, executed, and analyzed** a single-cell multi-omics study of **18 samples** and **+100,000 cells** across **6 time points**, covering mapping, QC, differential analyses, time-course modeling, and regulatory network inference to uncover mechanisms of cellular memory in neuronal systems.
- **Generated and analyzed +10 large-scale genomic datasets** incorporated into **5 scientific publications**.
- **Co-led** a collaborative project resulting in a **co-first-author *Nature*** (2021) publication (> **200 citations**), integrating scRNA-seq, scATAC-seq, and 3D genome architecture across brain cell types.
- **Developed** robust, reusable **analytical workflows** adopted across the group, improving reproducibility and standardizing analysis practices.
- **Awarded a competitive** DFG-funded **PhD scholarship,** a collaborative venture between Humboldt and Duke Universities that recognized academic excellence and research capabilities.
- **Awarded an outstanding research contributions prize** and a competitive **travel fellowship**; delivered **2 selected talks** and presented **5 posters** at international conferences.
- **Mentoring 2 researchers** in computational analysis strategies and served as **co-reviewer for 5 journals**.

Data Scientist | Bioinformatics & Genomics **Feb 2017 – Aug 2018**
New York University Medical School (NYUMS) *New York, US*

- **Led** a project investigating MED12 mutations in chronic lymphocytic leukemia, generating and analyzing **+20 genomic ChIP-seq, RNA-seq, and ATAC-seq datasets** from primary patient blood samples and cell lines.
- **Delivered** computational analysis of **+10 ChIP-seq and ATAC-seq datasets** for studies on inflammation and leukemic transformation contributing to **two high impact publications** *Nature Immunology* and *Cell Stem Cell.*

Research Fellow | Molecular Biology & Genomics **Oct 2015– Sept 2016**
European Molecular Biology Laboratory (EMBL) *Rome, Italy*
Max-Planck-Institute for Molecular Genetics (MPIMG) *Berlin, Germany*

- Gained hands-on molecular biology experience bridging wet-lab and computational expertise across two leading European research institutes.

EDUCATION

PhD in Systems Biology • Humboldt University of Berlin • Berlin, Germany (*Summa cum laude*) • **2025**
MSc in Bioinformatics • Autonomous University of Barcelona • Barcelona, Spain • **2017**
BSc in Biochemistry and Molecular Biology • University of Córdoba • Córdoba, Spain • **2015**

SKILLS

Programming & Development: Python, R, Shell, Git, Docker.
Statistical Analysis & Machine Learning: Statistical modeling, time-series analysis, dimensionality reduction, unsupervised learning (clustering, PCA).
Core Data Stack: NumPy, Pandas, Scikit-learn.
Bioinformatics: Multi-omics analysis, single-cell analysis (Seurat, Scanpy), genomics pipelines (STAR, Bowtie2, MACS2, deepTools), regulatory network inference.
Project & Communication: Cross-functional teamwork, project planning, mentoring, scientific writing, peer review, conference presentation.
Experimental Background: Hands-on experience with genomic library prep and molecular biology techniques, ensuring deep understanding of data generation and quality requirements.
Languages: English (fluent), Spanish (native), German (intermediate).

📊 Full project details, publications, and additional materials available at: **red-moonx.github.io**